1. Sets

1. What is a Set?

In our daily lives, we use words like "bouquet" for flowers, "bunch" for keys, or "flock" for birds. In math, when we can clearly and definitely decide which objects belong in a group, we call that collection a Set.

- **How to tell if it's a Set:** If everyone agrees on what belongs in the group, it is a set. If it depends on someone's opinion (like "happy" or "brilliant"), it is not a set because those meanings change from person to person.
- **Real-Life Example:** "The planets in our solar system" is a set because everyone knows exactly which planets to list. But "The best students in class" is not a set because everyone might have a different opinion on who is "best".
- **Exam Tip:** Look out for "relative" words like brave, strong, rich, or happy. If you see these, it's not a set!

2. How to Write a Set

We follow a few simple rules when writing sets:

- **Use Capital Letters (A, B, C...)** to name the set.
- **Use Small Letters (a, b, c...)** for the members or "elements" inside the set.
- **The Symbol $\in$:** If 'a' is a member of set A, we write $a \in A$ (a belongs to A). If it is not, we write $a \notin A$ (a does not belong to A).
- **Standard Sets of Numbers you should know:**
    - N = {1, 2, 3, ...} (Natural numbers).
    - W = {0, 1, 2, 3, ...} (Whole numbers).
    - I = {..., -3, -2, -1, 0, 1, 2, ...} (Integers).

3. Methods of Writing Sets

There are two ways to describe a set:

I. Listing Method (Roster Method)

You list every element inside curly brackets { } and separate them with commas.

- **Rule:** Each element is written only once, even if it appears many times in a word.
- **Example:** The set of letters in the word "remember" is $A = \{r, e, m, b\}$.
- **Real-Life Example:** A grocery list is like a roster method set!

II. Rule Method (Set Builder Form)

Instead of listing everything, we describe the "rule" or property that the elements share.

- **Format:** $A = \{x | x \text{ is a property}\}$.

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- **How to read it:** "A is the set of all x such that x is...".
- **Example:** $B = \{x | x \text{ is a prime number between 1 and 10}\}$. In listing form, this would be $B = \{2, 3, 5, 7\}$.

4. Types of Sets

Sets come in different "sizes":

- **Singleton Set:** Has only one element.
  - **Example:** $A = \{2\}$ (The set of even prime numbers).
- **Empty Set (Null Set):** Has no elements. We write it as $\{\}$ or use the symbol ϕ (phi).
  - **Example:** The set of humans who are 500 years old.
- **Finite Set:** You can count the elements. It has an end.
  - **Example:** Days of the week.
- **Infinite Set:** The elements are unlimited and cannot be counted.
  - **Example:** The set of natural numbers $\{1, 2, 3, ...\}$.

5. Equal Sets

Two sets, A and B, are equal ($A = B$) if every element in A is also in B, and every element in B is also in A.

- **Important:** The order doesn't matter!
- **Example:** $A = \{l, i, s, t, e, n\}$ and $B = \{s, i, l, e, n, t\}$. These are equal because they have the exact same letters.

6. Venn Diagrams and Subsets

Venn Diagrams

A British logician named John Venn used closed shapes like circles or rectangles to show sets. This makes them very easy to see.

Subsets ($\subseteq$)

If every element of set B is also an element of set A, then B is a subset of A. We write it as $B \subseteq A$.

- **Diagram:** In a Venn diagram, the circle for B would be inside the circle for A.
- **Real-Life Example:** If set A is "All students in your school" and set B is "Students in your class," then B is a subset of A.

Must-Remember Rules for Subsets:

1. Every set is a subset of itself ($A \subseteq A$).
2. The Empty set is a subset of every set ($\phi \subseteq A$).

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7. Universal Set and Complement

Universal Set (U)

The "Big Set" that contains everything we are currently talking about. In diagrams, we usually show it as a Rectangle.

- **Example:** If we are talking about a cricket team, the "Universal Set" would be all the students in the school.

Complement of a Set (A')

The complement of A contains all the elements that are in the Universal set but NOT in A.

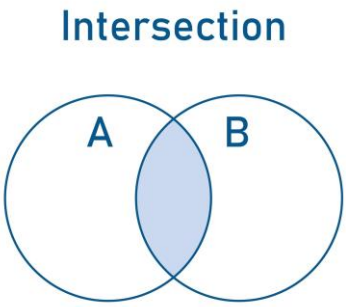
- **Property:** If you take the complement of a complement, you get the original set again! $(A')' = A$.

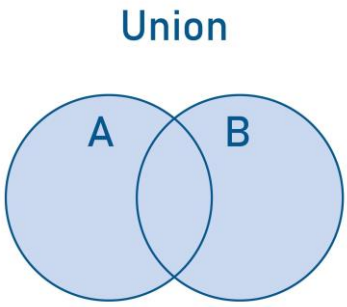
8. Operations on Sets (The "Action" Part)

I. Intersection ($\cap$)

The intersection of sets A and B is the set of elements that are common to both.

- **Symbol:** $\cap$ (looks like a cap).
- **Example:** If $A = \{1, 3, 5\}$ and $B = \{2, 3, 6\}$, then $A \cap B = \{3\}$ because 3 is in both.
- **Disjoint Sets:** If A and B have nothing in common, they are called Disjoint Sets. Their intersection is empty.





II. Union ($\cup$)

The union of A and B is a set containing all the elements from both sets.

- **Symbol:** $\cup$ (looks like a cup).
- **Rule:** Don't repeat common elements; just write them once.

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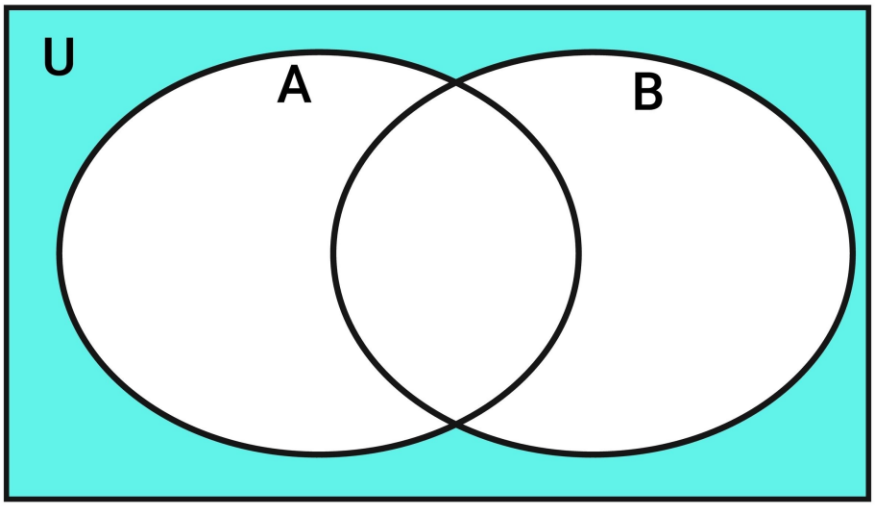
(A U B)



9. Number of Elements and The Master Formula

We write the number of elements in set A as $n(A)$.

The Formula:

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

Why do we use it?

When we add all elements of A and all elements of B, the common elements ($A \cap B$) get counted twice. We subtract them once so our total count is correct.

- **$n(A \cup B)$:** Total elements in both sets combined.
- **$n(A)$:** Elements in set A.
- **$n(B)$:** Elements in set B.
- **$n(A \cap B)$:** Elements they share.

Example Problem: In a class of 70 students, 45 like Cricket (A) and 52 like Kho-kho (B). All like at least one game.

- $n(A \cup B) = 70$
- $45 + 52 - n(A \cap B) = 70$
- $97 - n(A \cap B) = 70 \rightarrow n(A \cap B) = 27$ students like both.

Summary

- A Set is a well-defined collection.
- Listing method lists elements; Rule method describes them.

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- Subsets are smaller groups inside a larger group.
- Intersection ($\cap$) is what is shared; Union ($\cup$) is everyone combined.
- Use the Master Formula for word problems to avoid double-counting.

Key Terms

- **Element:** An object belonging to a set.
- **ϕ (Null Set):** A set with absolutely nothing in it.
- **Finite:** Something you can finish counting.
- **Complement (A'):** Everything in the Universal set except for A.

Exam-Oriented Questions

1. **Is "Rich people in the village" a set? Why?**
    - **Answer:** No. "Rich" is a relative term. The definition of rich varies from person to person, so it is not well-defined.
2. **Write the set of letters in the word 'MISSISSIPPI' using the listing method.**
    - **Answer:** $M = \{M, I, S, P\}$. (Remember: do not repeat letters!).
3. **If $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{4, 5, 6, 7, 8\}$, find $A \cap B$ and $A \cup B$.**
    - **Answer:** $A \cap B = \{4, 5, 6\}$ (shared items). $A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8\}$ (all items).
4. **What is a Singleton set? Give an example.**
    - **Answer:** A set with only one element. Example: $A = \{ \text{The even prime number} \}$ which is $\{2\}$.
5. **In a group of 100 people, 72 speak English and 43 speak French. Each person speaks at least one language. How many speak both?**
    - **Answer:** Use the formula $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.
        - $100 = 72 + 43 - n(A \cap B)$
        - $100 = 115 - n(A \cap B)$
        - $n(A \cap B) = 15$. So, 15 people speak both.